

Identifying Key Predictors of Sudden Cardiac Death

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Abstract

Sudden Cardiac Death, primarily due to cardiac arrhythmia, is a leading cause of mortality, responsible for approximately half of all deaths in cardiac disease. Left ventricular ejection fraction (LVEF) is almost exclusively used to identify patients who are most at risk for SCD and are candidates for the primary prevention of SCD, implantable cardiac defibrillators (ICDs). However, most cardiac arrests occur in patients with preserved or moderately reduced LVEF that are otherwise marked as low risk for SCD. This study aims to identify the most significant cardiac markers that contribute to risk identification for cardiac mortality and sudden cardiac death in patients with congestive heart failure in addition to LVEF. Factor analysis grouped significant clinical markers strongly associated with cardiac mortality into three dimensions: heart rate control, cardiac structure & function, and ventricular depolarization/repolarization duration. These domains align with established clinical risk factors and are a framework for understanding risk of sudden cardiac death. This study concludes that incorporating these additional variables beyond just the left ventricular ejection fraction can significantly improve the ability to predict mortality and guide prevention strategies in congestive heart failure patients.

Introduction & Literature Review

Sudden Cardiac Death (SCD) is a leading cause of mortality, resulting in over 300,000 deaths annually in the USA alone, and responsible for approximately half of all deaths in cardiac disease (Aro *et al.*, 2017). SCD is often referred to as a sudden death occurring within one hour of symptom onset when witnessed or an unexpected death from a cardiovascular cause where the person was seen alive in the past 24 hours in the case the death itself is not observed (Aro *et al.*, 2017). These sudden cardiac deaths usually occur as a result of cardiac arrhythmia. Currently, the primary prevention for SCD in patients with congestive heart failure (CHF) is the placement of an implantable cardioverter defibrillator (ICD). ICD's are recommended in patients with New York Heart Associate (NYHA) functional class II or III symptoms and left ventricle dysfunction with a left ventricle ejection fraction (LVEF) of $< 35\%$ (Shen *et al.*, 2017). A study exploring the efficacy

of ICD treatment found that in patients who are within NYHA class II or III and have heart failure with a reduced ejection fraction (HFrEF) with LVEF <35%, shock-only ICD therapy reduces overall mortality by 23% (Bardy *et al.*, 2005).

However, identification of SCD risk solely by LVEF has its limitations. In fact, most cardiac arrests occur among subjects with preserved or only moderately reduced LVEF who are otherwise classified as low risk just by using LVEF as a marker (Aro *et al.*, 2017). Despite the efficacy of ICD therapy in patients with systolic heart failure caused by coronary artery disease being well documented, the implantation of an ICD in patients that had heart failure not caused by ischemic heart disease did not provide an overall survival benefit (Kober *et al.*, 2016). ICD therapy is costly, and comes with the risk of post-implantation complications that affect quality of life. These include infections at the implantation site or inappropriate shocks and device malfunctions (Shen *et al.*, 2017). The need arises for the exploration of alternative markers to be used for better SCD risk stratification alongside LVEF. Previous studies have identified three categories of SCD risk: Clinical Risk Factors (HTN, Dyslipidemia, Atrial Fibrillation, Lifestyle/Dietary Factors), Electrocardiographic parameters (short or long QT syndromes, QRS intervals), and History of Coronary Heart Disease (Wong *et al.*, 2019).

This study addresses the following research questions. The first asks what latent dimensions underlie clinical and electrocardiogram (ECG) measures in CHF patients (RQ1) with the hypothesis that dimensionality reduction will reveal a small number of meaningful factors that summarize the variation in clinical and ECG measures. The second asks which variables are most significant for predicting sudden cardiac death beyond LVEF (RQ2), hypothesizing that multiple clinical, ECG, and lab values contribute to risk prediction of SCD. The third asks what the nature of the relationship between clinical heart failure classification and laboratory biomarkers is (RQ3) with a hypothesis that multiple biomarkers, specifically cardiac markers, significantly reflect the severity of heart failure. Finally, the fourth asks what is the relationship between ECG rhythm and implantational devices (RQ4) hypothesizing patients with implantable devices exhibit a shift in rhythm dependency.

Methods

The dataset used from this analysis came from the MUerte Subita en Insuficiencia Cardiaca (MUSIC) study with the goal of assessing risk predictors of cardiac mortality in patients with CHF. The study consisted of a population of 992 CHF patients who underwent baseline testing including an echocardiogram (to check for how well the heart is pumping), chest X-ray, 24 hour Holter ECG monitoring, clinical evaluation, and laboratory tests that checked for cardiac values such as troponin or NT-proBNP. From these measurements, the researchers created a risk profile for each patient and assigned risk points (MUSIC risk score). The patients were followed for over 44 months

with check-ups every 6 months to evaluate who died and if the baseline risk profile predicted it. During each of the check-ups, the researchers recorded major outcomes such as heart transplants, non-heart related deaths, survival, sudden cardiac death (SCD), or other heart-related deaths.

One of the statistical methods that was employed to analyze our data was PCA due to the size of the dataset. The study focused on reducing the dimensions of the dataset to the number of significant variables that could contribute to understanding the primary risk factors contributing to cardiac mortality in patients with chronic heart failure. Factor Analysis was implemented to understand the latent clinical dimensions present in the dataset relating to risk of cardiac mortality. Correlations of each variable were checked and the ones with high coefficients (>0.8) were eliminated to prevent multicollinearity. KMO, Bartlett's test, and Chronbach's alpha were used to determine the factor reliability. This study used a significance level of $\alpha = 0.05$ (two-tailed) for all statistical methods.

Clinical Classification and Laboratory Tests were created to assess the relationship between clinical heart failure classification and laboratory biomarkers by using Canonical Correlation Analysis (CCA). Clinical Classification contains two variables: New York Heart Association (NYHA) class and HF etiology-Diagnosis. The Laboratory Tests set includes Troponin, Creatinine, Potassium, Glucose, Alanine Aminotransferase or Glutamic Pyruvic Transaminase, Total Cholesterol, Thyroid Stimulating Hormone, Hemoglobin, Albumin, High-density lipoprotein cholesterol, and Low Density Lipoprotein Cholesterol. ECG rhythm (categorical) and prior implantable device (categorical) were cross-tabulated, and independence was tested with Pearson's chi-square. Correspondence analysis (CA) was then applied to the contingency table to produce a low-dimensional map in which proximity reflects over-representation relative to independence. The analysis reports the chi-square statistic, degrees of freedom, p-value, percent inertia by dimension, and the categories with the highest contributions to Dimension 1. Guided by the CA geometry, a clinically coherent indicator of paced versus non-paced rhythm was derived for use in the latent-dimension (RQ1) and prediction (RQ2) models.

All analyses in this study were completed using R Studio (v. 2025.05.1+513) and the following packages: REdaS (v. 0.9.4), yacca (v. 1.4-2), fmsb (v.0.7.6), psych (v. 2.5.6), ca (0.71.1), vcd (1.2-13)..

Results

As a descriptive check prior to PCA/FA, ECG rhythm and prior implantable device status showed a very strong association (Pearson's $X^2=778.12$, $df=21$, $p<2.2 \times 10^{-16}$; $n=992$). Row percentages indicated that patients without a prior device (code 0) were almost entirely in non-paced rhythms (columns 0–1; $\approx 99\%$), whereas device-positive categories (e.g., codes 1–4) concentrated in the paced rhythm category (column 3; $\approx 65\% \text{--} \approx 89\%$). Correspondence analysis

(CA) summarized this structure on a single dominant axis: Dimension 1 explained $\approx 98\%$ of the chi-square inertia (Dim 2 $\approx 1.8\%$), with the largest contributions from device codes 3–4–1–2 on the row side and rhythm code 3 on the column side. In the CA map, device-negative aligned with non-paced rhythms (left of the origin), and device-positive aligned with paced rhythm (right of the origin), matching clinical expectations. The shaded mosaic in Figure 1 shows standardized residuals: blue cells are above independence expectations and red below. “No prior device” with rhythm 0 is strongly over-represented, while device codes 2–4 are over-represented with non-zero rhythms (especially rhythm 3). Correspondence analysis of the same table (Figure 2) yields a two-dimensional map where Dimension 1 explains $\approx 98.0\%$ of inertia and cleanly separates the “no device + rhythm 0” cluster (left/near origin) from device codes 2–4 aligned with non-zero rhythms (right), reinforcing the mosaic’s pattern.

Kaiser-Meyer-Olkin Test, Bartlett’s Test of Sphericity, and Cronbach Alpha score were calculated to determine the sampling adequacy. In Table 1, KMO resulted in 0.57, Bartlett’s Test of Sphericity resulted in 9585.61 of the test statistic ($df = 861$, $p < 2.22e-16$), and the Cronbach Alpha score resulted in 0.117. Though Cronbach Alpha score of 0.117 was relatively low, which indicates poor internal consistency, the KMO score greater than 0.5 indicates a mediocre sampling adequacy, and the p-value in Bartlett’s Test of Sphericity less than 0.05 indicates that there were adequate differences and variances that could be separated into different factors or principal components. Overall, variables within the dataset had sufficient sample size and correlated enough for principal component analysis as well as factor analysis.

Principal component analysis was used to reduce the number of dimensionality of the dataset. A solution with twenty-two principal components was identified as optimal, capturing over 80% of the total variance (Figure 3) with a variable loadings cutoff at 0.4. Thus, the data could be reduced to twenty-two dimensions while still retaining the most information. Based on the scree plot (Figure 4), the ‘elbow’ suggests to keep 3-4 components, however, minimal components only explained 24% of the variance within the model. In order to incorporate a higher degree of variance, more components were included to capture pathways to cardiac mortality. Although only 15 components are above an eigenvalue of 1, 22 components were kept to explain 80% of variation within our data.

In Figure 3, the first principal component only captures 9% of the total variance in the dataset. PC1 is heavily influenced by age (0.58), QT interval (0.61), minimum RR intervals (0.51), average RR intervals (0.51), urea (0.47), and creatinine (0.41), which indicates a pattern of cardiac abnormalities and kidney function. PC2 is heavily influenced by LVEF (0.58), systolic blood pressure (0.58), septal thickness (0.58), and LV posterior wall thickness (0.54), which captures cardiac structure and function. PC3 is heavily influenced by average RR intervals (-0.70) and minimum RR intervals (-0.45). Overall, the most influential predictors include average RR intervals, QT intervals, LVEF, systolic blood pressure, and age. Factor Analysis condensed

numerous cardiac measures into three dimensions: Heart Rate Control, Cardiac Structure & Function, and Ventricular Depolarization and Repolarization Duration. A cutoff of 0.40 was used for the variable loadings. The variables that contributed to these dimensions are shown in Table 2. These factors together explained 57% of the variation in the data (Table 3).

The canonical correlation value of Canonical Variate (CV) is 0.2335, indicating a positive but weak relationship between Clinical Classification and Laboratory Tests, so Laboratory Tests were positively but weakly related to Clinical Classification. Bartlett's Chi-Squared Test shows the canonical correlation for the first variate was significant and the relationship between two sets was not likely due to random chance because of the first variate with a p-value less than 0.05. In Table 4, Creatinine had the highest absolute value among variables in Laboratory Tests, indicating Creatinine contributes most to the formation of Laboratory Tests. NYHA class in the Clinical Classification set strongly contributes to CV, but HF etiology - Diagnosis with a negative value showed an opposite contribution. The longest and black bars of Creatinine and NYHA class and the white short bar of HF etiology – Diagnosis in the Helio Plot (Figure 5) also supported this finding.

Discussion

Electrophysiological Activity & Device Status

The strong association between ECG rhythm and implantable cardiac device status justifies recoding ECG rhythm into a binary paced vs non-paced indicator and including device status in the subsequent latent-dimension (RQ1) and prediction (RQ2) analyses. Ventricular Depolarization and Repolarization Duration emerged as a key factor in the data which the rationale for implantable cardioverter-defibrillators (ICDs) as a primary intervention to prevent fatal arrhythmias. Correspondence analysis demonstrated that patients without a prior implantable device almost entirely exhibited non-paced rhythms while patients with implantable devices grouped into paced rhythms, supporting clinical expectations and improvement with intervention. PCA identified QT prolongation as a significant predictor in cardiac mortality as it represents ventricular repolarization/depolarization. QT interval abnormalities and changes in ventricular electrical activity can lead to cardiac death in patients with or even without genetic predispositions.

Age, Cardiac Structure, and Function

As an individual ages, the heart becomes more fragile and can begin to wear out leading to higher chances of cardiac diseases. Age, LVEF, and systolic blood pressure were consistently noted as significant predictors of cardiac mortality. PCA highlighted age as a significant contributor to the first component, reflecting the critical effect age has on predisposing patients to cardiac death.

Factor analysis identified cardiac structure and function as a significant dimension among individuals with CHF which align with established triggers of sudden cardiac death, including cardiac arrest from acute electrical or mechanical failure of the ventricles in patients with remodeled or fibrotic ventricles (ventricles that have changed in shape and function, or have developed scar tissue due to cardiac damage or disease). NYHA class, a way of classifying heart failure, emerged as a significant clinical correlate in CCA. Consistent with prior studies, high NYHA class rating, indicating worsening cardiac function and symptoms, correlates with higher risk of sudden cardiac death in CHF.

Systolic blood pressure measures the pressure the heart squeezes. If the pressure is too high, this can overwork the heart and cause issues. On the other hand, if the pressure is low, organs are not receiving the appropriate amount of blood to sustain them. Both FA and PCA underscored the importance of blood pressure control in reducing risk of SCD. LVEF emerged as a significant variable across the analysis, supporting its use as a clinical predictor of SCD risk, particularly in identifying patients that are candidates for ICD therapy (Shen *et al.*, 2017). However, most cardiac arrests occur among subjects with preserved or only moderately reduced LVEF (Aro *et al.*, 2017). This underscores the need for additional variables that combined with LV systolic function enhance the risk stratification of SCD in patients with CHF.

These results highlight the importance of early screening and monitoring, with increased access to echocardiograms and personalized clinical interventions, especially as people age. These interventions may include patient counseling and support for lifestyle and dietary changes to prevent progression of heart failure and reduce mortality.

Heart Rate Control

Heart rate measures, including average RR intervals, QT interval, and ventricular depolarization/repolarization, consistently emerged as strong predictors. PCA and FA demonstrated that these variables, mainly Average RR, are important to the dimension of Heart Rate Control. Average RR intervals can determine cardiac mortality based on how the heart beats in certain conditions. This dimension underscores the clinical value of rate and rhythm management through medical therapy, device-based interventions, and lifestyle modifications such as exercise and diet. Additionally, the significance of this dimension supports the notion that both cardiac electrical and structural activity significantly contribute to cardiac mortality.

Multisystemic Markers

CCA and PCA results indicated that SCD can be multi-systematic and beyond just cardiac function. The organ systems cannot function individually; oftentimes one organ failure compromises multiple organs. Chronic Kidney Disease (CKD) is a common comorbidity amongst

individuals with CHF. Since creatinine and urea, lab values used to assess kidney function, contributed to the variate, it indicates that when determining the classification of cardiac function, not only cardiac markers should be focused on, but also other biomarkers such as inflammation and metabolism should be considered. Across all methods, heart structure, ventricular electrical activity, and heart rate measures consistently emerged as important predictors of SCD risk. PCA results (Figure 2) highlighted QT intervals, ejection fraction, and ventricular dimensions, while factor analysis confirmed their clustering into clinically relevant factors (Table 2 and Table 3). CCA and PCA further extended these results by linking clinical classifications to laboratory markers, notably creatinine and urea. These findings demonstrate that SCD is multifactorial, driven not only by structural abnormalities or cardiac electrical activity, but also by other systemic conditions such as renal impairment.

Limitations & Future Directions

In this research study, 11 patients were lost to follow up meaning that their outcomes were unknown in terms of if the patients survived or died due to cardiac or non cardiac reasons. The data from these patients could have been statistically meaningful, affecting the outcome of our analysis. Furthermore, many patients were removed from the analysis due to incomplete or missing data. Therefore, the outcome of our data could be biased towards those that had complete documentation. Finally, the data is based on patients from eight different Spanish University Hospitals from the MUSIC study which may introduce selection bias. There is a possibility that the data from this sample is not representative of a general population with congestive heart failure. The setting in which the research study occurred at specialized HF clinics from Spanish university hospitals, and the treatment may have differed from those in community hospitals in regards to access to care, socioeconomic background, and disease severity.

Another key limitation included potential measurement and interpretation bias during data collection. A few examples of these include clinical scales such as the New York Heart Association (NYHA) functional class. The clinical scale relies on the reporting from patients and physician judgement, lending to potential interpretation bias from the provider and the patient. Echocardiographic measurements such as left ventricular ejection fraction (LVEF) are operator dependent. Each operator had differences in training, equipment, and interpretation guidelines, thus leading to possible variations across the patient population. Since LVEF was a vital comparator for this research study, variations in the measurement and interpretation could have influenced the classification of heart failure severity.

Lastly, results from PCA and factor analysis were sensitive to the number of components that we chose to keep. Selecting too few could have led to missing important variables, while selecting too many may have introduced noise and less meaningful variables which could have impacted our interpretation of which factors are more strongly associated with sudden cardiac death risk (SCD).

While our analysis identified clinically relevant dimensions such as cardiac structure, ventricular repolarization, and heart rate control, these findings should be interpreted with caution.

Further research can be done to validate if the risk factors hold true across populations with different demographics and socioeconomic status. These studies would strengthen the generalizability of the findings, and to ensure that the risk prediction tools are accurate across the demographic groups. In addition, studies could be done to understand if there's a lifestyle or genetic component that contributes to cardiac risk. Expanding research across a larger population size can help refine the risk prediction tools to be applied globally. Future work should also focus on developing and validating a clinical risk scoring tool that combines these multiple dimensions. Additional research should also explore how lifestyle and genetic factors interact with these physiological markers to influence sudden cardiac death risk.

Future research could also be done to explore the relationship between genetic predispositions, family history, and lifestyle factors such as exercise, diet, and smoking and laboratory and clinical markers and how that influences sudden cardiac death risk. Understanding this relationship could uncover additional pathways of opportunities for early intervention. Lastly, with the rise of machine learning and artificial intelligence, machine learning methods could be used to develop predictive models that provide better accuracy than traditional statistical techniques. This could help identify patients who are at high risk for sudden cardiac death and help with preventive strategies.

Conclusion

In conclusion, this study demonstrates that sudden cardiac death in patients with congestive heart failure (CHF) is best predicted by a combination of heart rate control, structural function, and electrical activity, rather than ejection fraction alone. By moving toward a more holistic assessment, clinicians can better identify high-risk patients and provide timely interventions such as defibrillator therapy, medical optimization, and lifestyle modifications. Ultimately, these findings can improve patient outcomes and reduce the burden of sudden cardiac death in society.

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Tables & Figures

Table 1. Factorability Scores

Kaiser-Meyer-Oskin	Overall MSA: 0.57
Bartlett's Test of Sphericity	$X^2 = 7625.26$, $df = 820$, $p\text{-value} < 2.22e-16$
Cronbach Alpha	0.117

Table 2. Factor Analysis Results and Loadings

	Heart Rate Control	Cardiac Structure & Function	Ventricular Depolarization & Repolarization
Average RR (ms)	0.988		
Minimum RR (ms)	0.515		
Average RR (ms/L)	0.690		
Left Ventricular Ejection Fraction		0.816	
Septal Thickness (mm)		0.530	
Left Ventricle End Diastolic Diameter (mm)		-0.800	
QRS Duration (ms)			0.631
QT interval (ms)	0.657		<u>0.743</u>
QT corrected			0.958
Systolic Blood Pressure (mmHg)		0.434	
Left Ventricular Posterior Wall Thickness (mm)		0.458	

Table 3. Factor Analysis Results

	Heart Rate Control	Cardiac Structure & Function	Ventricular Depolarization & Repolarization
Sum of Squared Factor Loadings	2.225	2.062	2.003
Proportion Variance	0.202	0.187	0.182
Cumulative Variance	0.202	0.390	0.572

Table 4. Canonical Variate Coefficients (CV)

X Variables	CV
Alanine Aminotransferase (ALT) or Glutamic Pyruvic Transaminase (GPT) (IU/L)	-0.00122
Troponin (ng/mL)	-0.00313
Creatinine (mol/L)	0.0255
Potassium (mEq/L)	0.000226
Glucose (mmol/L)	0.00190
Thyroid Stimulating Hormone (TSH) (mIU/L)	-0.000529
Hemoglobin (g/L)	0.00265
Total Cholesterol (mmol/L)	-0.00117
Albumin (g/L)	-0.000129
Y Variables	CV
New York Heart Association (NYHA) class	2.36
HF etiology - Diagnosis	-0.0877

Figure 1. Mosaic Plot of Prior Implantable Device vs. Electrocardiogram (ECG) Rhythm

Prior implantable device × ECG rhythm

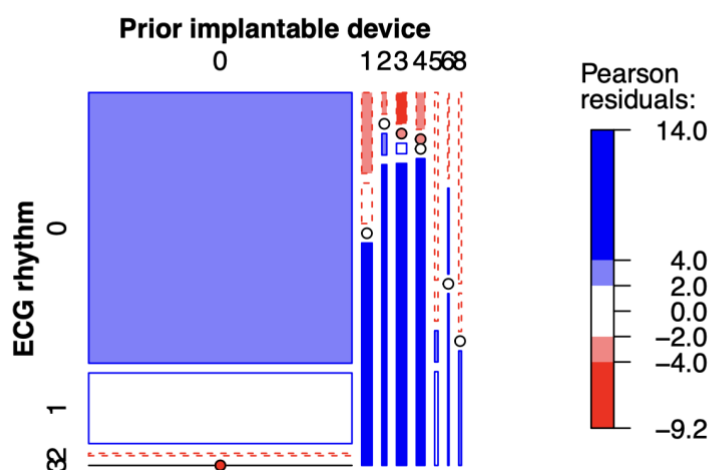


Figure 2. Correspondence Analysis Map for ECG Rhythm (Rows) and Prior Implantable Device (Columns)

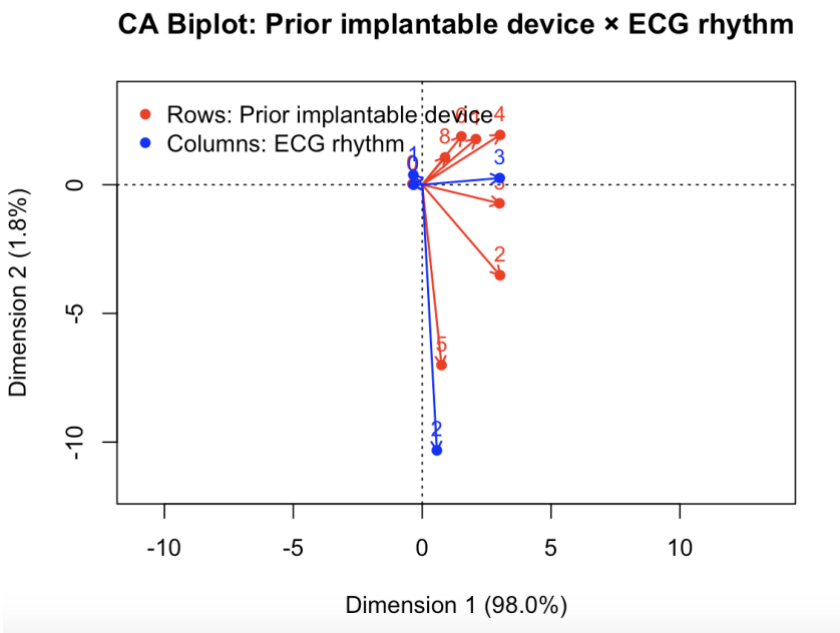


Figure 3. Principal Component Analysis Results

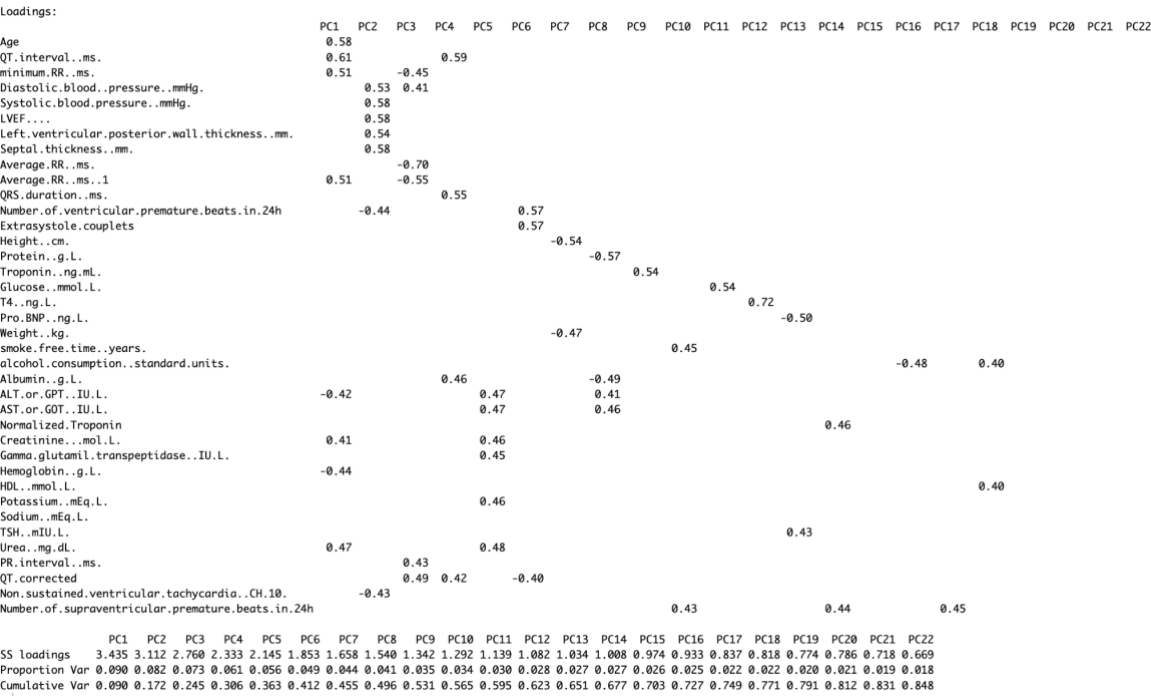


Figure 4. Scree Plot for Principal Component Analysis

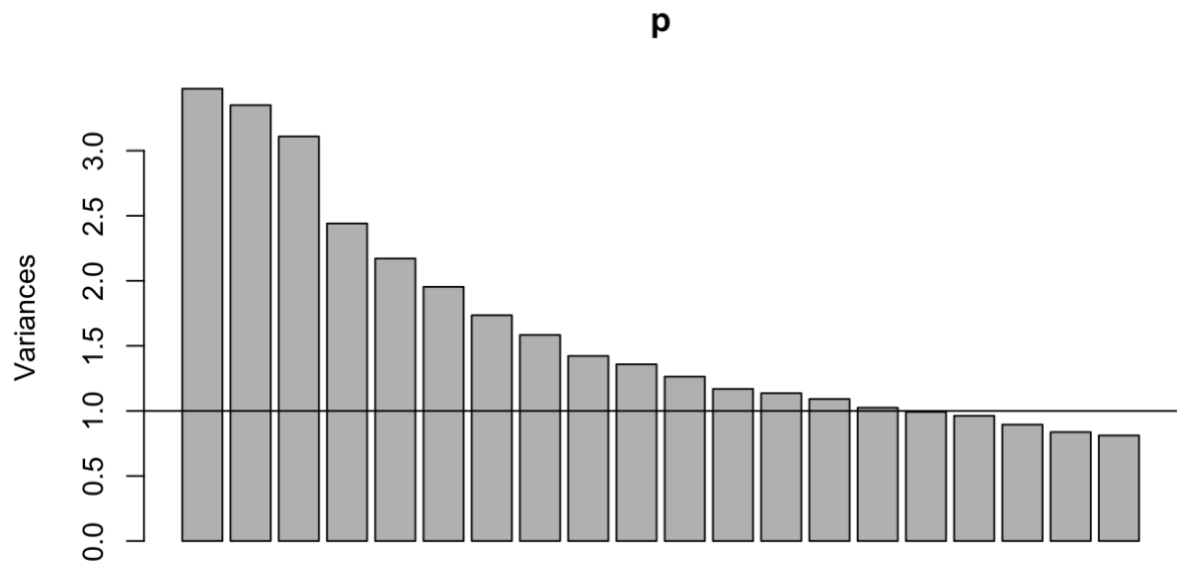


Figure 5. Clinical-Laboratory Relationship Helio Plot

